

DRY-RUN GUIDE | 06\_suffix\_count.java

Source: java\2nd\_belt\06\_suffix\_count.java

#### CORE BEHAVIOR

Counts repeated copies of the suffix at the very end of a token, moving left by suffix length after each match; it does not count arbitrary occurrences.

#### REPRESENTATIVE INPUT / CALL

String 'ababab', suffix 'ab'.

#### STEP TRACE

Trailing 'ab' matches (count 1); preceding 'ab' matches (2); preceding 'ab' matches (3); next window is too short.

#### EXPECTED OUTPUT / RETURN

'3'

#### ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Given a string and a suffix, count how many times the suffix appears at
004 | the end of the string.
005 |
006 | Input:
007 | A string S and a suffix X.
008 |
009 | Output:
010 | Print the number of times X appears as a suffix of S.
011 | */
012 |
013 | import java.util.Scanner;
014 |
015 | public class SuffixCount {
016 |     public static void main(String[] args) {
017 |         Scanner sc = new Scanner(System.in);
018 |         String s = sc.next();
019 |         String suffix = sc.next();
020 |
021 |         int count = 0;
022 |         int start = s.length();
023 |
024 |         while (start >= suffix.length()) {
025 |             String current = s.substring(start - suffix.length(), start);
026 |             if (current.equals(suffix)) {
027 |                 count++;
028 |                 start -= suffix.length();
029 |             } else {
030 |                 break;
031 |             }
032 |         }
033 |
034 |         System.out.println(count);
035 |         sc.close();
036 |     }
```

037 | }  
038 |